

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T04 · Powers, Roots and Standard Form

With Answers

Questions with full worked solutions

■ 67 pages

FAMILY MAP

Powers Roots Standard Form

F01	Convert and compare standard form	14 Q 25 marks
F02	Calculate and model in standard form	12 Q 28 marks
F03	Simplify numerical index expressions	6 Q 10 marks

Question 24

4MA1-2H Q12 (a)

ORIGINAL QUESTION UNIT 1

2 marks

12 $2^7 \times 4^5 = 4^x$

(a) Calculate the value of x

$$x = \dots\dots\dots$$

(2)

Worked solution 24

Solve an index equation

MODULAR UNIT 1

2 marks

(a) – Solve an index equation

Write all terms in base 2

$$2^7 \cdot 4^5 = 2^7 \cdot (2^2)^5 = 2^{17}, \quad 4^x = (2^2)^x = 2^{2x}.$$

Equate exponents

$$2x = 17 \Rightarrow x = 8.5.$$

Final answer

$$x = 8.5$$

Question 08

4MA1-1H Q15

ORIGINAL QUESTION

3 marks

15 (a) $\sqrt{2} \div \frac{8^3}{16^{\frac{3}{2}}} = 2^n$

Work out the value of n
Show your working clearly.

$n = \dots\dots\dots$
(3)

Worked Solution 08

Numerical indices

MODULAR UNIT 1

3 marks

Write every term as a power of 2

$$\sqrt{2} = 2^{1/2}, \quad 8^3 = 2^9, \quad 16^{3/2} = 2^6.$$

Subtract indices

$$2^{1/2} \div 2^{9-6} = 2^{1/2-3} = 2^{-5/2}.$$

Final answer

$$n = -\frac{5}{2} = -2.5$$

1 (a) Simplify $e^8 \div e^2$

(1)

Worked solution

November 2021 | Question 1 | 1 marks

Family: Simplify numerical index expressions | Variant: Multiply or divide powers with a common base

METHOD / WORKING

$$1. (a) e^8 \div e^2 = e^6.$$

FINAL ANSWER

$$(a)e^6$$

- 1 (a) Write down the value of m , given that $3^4 \times 3^5 = 3^m$

$m = \dots\dots\dots$

(1)

Worked solution

November 2021 | Question 1 | 1 marks

Family: Simplify numerical index expressions | Variant: Multiply or divide powers with a common base

METHOD / WORKING

1. (a) $3^4 \times 3^5 = 3^9$, so $m = 9$.

FINAL ANSWER

 $m=9, n=21, p=4$

(b) Write down the value of n , given that $(5^3)^7 = 5^n$

$n = \dots\dots\dots$
(1)

Worked solution

November 2021 | Question 1 | 1 marks

Family: Simplify numerical index expressions | Variant: Power of a power or solve a numerical index relation

METHOD / WORKING

1. (b) $(5^3)^7 = 5^{21}$, so $n = 21$.

FINAL ANSWER

 $m=9, n=21, p=4$

(c) Find the value of p , given that $\frac{7^8 \times 7^2}{7^p} = 7^6$

$p = \dots\dots\dots$
(2)

Worked solution

November 2021 | Question 1 | 2 marks

Family: Simplify numerical index expressions | Variant: Power of a power or solve a numerical index relation

METHOD / WORKING

1. (c)

$$2. \frac{7^8 \times 7^2}{7^p} = 7^{10-p} = 7^6$$

3. So $10 - p = 6$, hence $p = 4$.4. Answer: $m = 9$, $n = 21$, $p = 4$

FINAL ANSWER

m=9, n=21, p=4

8 (a) Write 520 million in standard form.

.....
(1)

(b) Write 8.79×10^{-5} as an ordinary number.

.....
(1)

Worked Solution 06

Standard form: two conversions

UNIT 2**2 marks****Large number**

$$520\,000\,000 = 5.2 \times 10^8.$$

Small number

$$8.79 \times 10^{-5} = 0.0000879.$$

Final answer

$$5.2 \times 10^8, \quad 0.0000879$$

- (c) Work out $(5 \times 10^{42}) \times (7 \times 10^{-180})$
Give your answer in standard form.

.....
(2)

Worked Solution 07

Multiplication in standard form

UNIT 2
2 marks**Multiply coefficients and powers**

$$(5 \times 10^{42})(7 \times 10^{-180}) = 35 \times 10^{-138} = 3.5 \times 10^{-137}.$$

Final answer

$$3.5 \times 10^{-137}$$

Question 05

4MA1-1H Q9

ORIGINAL QUESTION

3 marks

- 9 The table gives the amount of rice produced by each of two countries in 2020

Country	Amount of rice (tonnes)
Indonesia	3.5×10^7
Argentina	8.2×10^5

- (a) Write 3.5×10^7 as an ordinary number.

.....
(1)

In 2020, Japan produced 6 780 000 more tonnes of rice than Argentina.

- (b) Work out the amount of rice Japan produced in 2020
Give your answer in standard form.

..... tonnes
(2)

Worked Solution 05

Standard form

MODULAR UNIT 2

3 marks

Ordinary number

$$3.5 \times 10^7 = 35\,000\,000.$$

Addition

$$8.2 \times 10^5 + 6\,780\,000 = 7\,600\,000 = 7.6 \times 10^6.$$

Final answer

$$35\,000\,000; \quad 7.6 \times 10^6 \text{ tonnes}$$

Question 06

4MA1-1H Q9 (a)

ORIGINAL QUESTION UNIT 2

1 marks

- 9 (a) Write 5.87×10^{-4} as an ordinary number.

.....
(1)

Worked solution 06

Ordinary number

MODULAR UNIT 2

1 marks

(a) – Ordinary number

Move the decimal four places left

$$5.87 \times 10^{-4} = 0.000587.$$

Final answer

0.000587

Question 07

4MA1-1H Q9 (b)

ORIGINAL QUESTION UNIT 2

1 marks

(b) Write 84 000 000 in standard form.

.....
(1)

Worked solution 07

Standard form

MODULAR UNIT 2

1 marks

(b) – Standard form

Write one non-zero digit first

$$84000000 = 8.4 \times 10^7.$$

Final answer

$$8.4 \times 10^7$$

Question 08

4MA1-1H Q9 (c)

ORIGINAL QUESTION UNIT 2

2 marks

The number of neurons in a human brain is 8.5×10^{10}
The number of neurons in a monkey brain is 1.47×10^9

The number of neurons in a human brain is $K \times$ the number of neurons in a monkey brain.

- (c) Work out the value of K
Give your answer correct to one decimal place.

$K =$
(2)

Worked solution 08

Standard-form ratio

MODULAR UNIT 2

2 marks

(c) – Standard-form ratio

Form the ratio

$$K = \frac{8.5 \times 10^{10}}{1.47 \times 10^9} = \frac{85}{1.47}.$$

Evaluate

$$K = 57.823... \approx 57.8.$$

Final answer

$$K = 57.8$$

Question 13

4MA1-1H Q15

ORIGINAL QUESTION

3 marks

- (b) Find 4% of 4.5×10^{157}
Give your answer in standard form.

.....
(3)

Worked Solution 13

Percentage of a standard-form number

MODULAR UNIT 2

3 marks

Change 4 percent to a multiplier

$$4\% = 0.04 = 4 \times 10^{-2}.$$

Multiply and normalize

$$(4 \times 10^{-2})(4.5 \times 10^{157}) = 18 \times 10^{155} = 1.8 \times 10^{156}.$$

Final answer

$$1.8 \times 10^{156}$$

Question 20

4MA1-2H Q6

ORIGINAL QUESTION

2 marks

6 (a) Write 76 000 000 in standard form.

.....
(1)

(b) Write 5.4×10^{-4} as an ordinary number.

.....
(1)

Worked Solution 20

Standard form conversions

MODULAR UNIT 2

2 marks

Large ordinary number

$$76\,000\,000 = 7.6 \times 10^7.$$

Small ordinary number

$$5.4 \times 10^{-4} = 0.00054.$$

Final answer

$$7.6 \times 10^7, \quad 0.00054$$

Original question 08

4MA1-1H Q9 M01

MODULAR UNIT 2

1 marks

- 9 (a) Write 8.4×10^{-5} as an ordinary number.

.....
(1)

Worked solution 08

Convert standard form to an ordinary number

MODULAR UNIT 2

1 marks

M01 – Convert standard form to an ordinary number**Move the decimal four places left**

$$8.4 \times 10^{-5} = 0.000084.$$

Final answer

0.000084

Original question 09

4MA1-1H Q9 M02

MODULAR UNIT 2

2 marks

- (b) Work out $(6.5 \times 10^{-40}) \times (8 \times 10^{185})$
Give your answer in standard form.

.....
(2)

Worked solution 09

Multiply in standard form

MODULAR UNIT 2

2 marks

M02 – Multiply in standard form**Multiply the numbers and powers**

$$(6.5 \times 10^{-40})(8 \times 10^{185}) = 52 \times 10^{145}.$$

Normalise

$$52 \times 10^{145} = 5.2 \times 10^{146}.$$

Final answer

$$5.2 \times 10^{146}$$

Original question 09

4MA1-1H Q8 Q8(a)

ORIGINAL QUESTION**1 marks**

- 8 (a) Write 0.000 089 in standard form.

.....
(1)

Worked solution 09

Chapter: Powers, Roots & Standard Form

MODULAR UNIT 2

1 marks

Q8(a) – Chapter: Powers, Roots & Standard Form

Family: Convert and compare standard form • Variant: Ordinary number to standard form

Method

$$0.000089 = 8.9 \times 10^{-5}.$$

*Why: Move the decimal point five places right to obtain a number between 1 and 10.***Final answer****standard form:** 8.9×10^{-5}

Original question 10

4MA1-1H Q8 Q8(b)

ORIGINAL QUESTION**1 marks**

(b) Write 8.34×10^4 as an ordinary number.

.....
(1)

Worked solution 10

Chapter: Powers, Roots & Standard Form

MODULAR UNIT 2

1 marks

Q8(b) – Chapter: Powers, Roots & Standard Form

Family: Convert and compare standard form • Variant: Standard form to ordinary number

Method

$$8.34 \times 10^4 = 8.34 \times 10000 = 83400.$$

*Why: A positive power of 10 moves the decimal point four places right.***Final answer****ordinary number:** 83400

Original question 07

4MA1-1H Q8

ORIGINAL QUESTION**2 marks****8**

$$a = 4.2 \times 10^{-24}$$

$$b = 3 \times 10^{145}$$

Work out the value of $a \times b$

Give your answer in standard form.

Worked solution 07

Chapter: Powers, Roots & Standard Form

MODULAR UNIT 2

2 marks

Q8 – Chapter: Powers, Roots & Standard Form

Family: Calculate and model in standard form • Variant: Multiply or divide standard-form numbers

Method

$$(4.2 \times 10^{-24})(3 \times 10^{145}) = 12.6 \times 10^{121}.$$

*Why: Multiply coefficients and add powers of ten.***Method**

$$12.6 \times 10^{121} = 1.26 \times 10^{122}.$$

*Why: Move the decimal one place and increase the exponent by one.***Final answer****standard form:** 1.26×10^{122}

Original question 26

4MA1-2H Q12

ORIGINAL QUESTION

3 marks

12 $a = 6 \times 10^{40}$

Work out the value of a^3

Give your answer in standard form.

Worked solution 26

Chapter: Powers, Roots & Standard Form

MODULAR UNIT 2

3 marks

Q12 – Chapter: Powers, Roots & Standard Form

Family: Calculate and model in standard form • Variant: Raise a standard-form quantity to a power

Method

$$(6 \times 10^{40})^3 = 6^3 \times 10^{120} = 216 \times 10^{120}.$$

*Why: Cube the coefficient and multiply the exponent by 3.***Method**

$$216 \times 10^{120} = 2.16 \times 10^{122}.$$

*Why: Normalize the coefficient to standard form.***Method**Therefore the value is 2.16×10^{122} .*Why: State the final answer.***Final answer****standard form:** 2.16×10^{122}

6 (a) Write 7.8×10^{-4} as an ordinary number.

(1)

Worked solution

January 2020 | Question 6 | 1 marks

Family: Convert and compare standard form • Variant: Standard form to ordinary number

Method / working

1. Move the decimal four places left for 10^{40} .

Final answer: 0.00078

(b) Work out $\frac{5.6 \times 10^4 + 7 \times 10^3}{2.8 \times 10^{-3}}$

Give your answer in standard form.

(2)

Worked solution

January 2020 | Question 6 | 2 marks

Family: Calculate and model in standard form • Variant: Add or subtract standard-form numbers

Method / working

1. $5.6 \times 10^4 + 7 \times 10^3 = 63000$.

2. $63000 / (2.8 \times 10^3) = 2.25 \times 10^7$.

Final answer: 2.25×10^7

6 (a) Write 2 840 000 000 in standard form.

(1)

Worked solution

January 2021 | Question 6 | 1 marks

Family: Convert and compare standard form • Variant: Ordinary number to standard form

Method / working

1. Move the decimal point nine places to write 2840000000 in standard form.

Final answer: 2.84×10^9

(b) Write 2.5×10^{-4} as an ordinary number.

(1)

Worked solution

January 2021 | Question 6 | 1 marks

Family: Convert and compare standard form • Variant: Standard form to ordinary number

Method / working

1. 2.5×10^{-4} means move the decimal four places left.

Final answer: 0.00025

(b) Work out $\frac{9.6 \times 10^{141} + 6.4 \times 10^{140}}{3.2 \times 10^{16}}$

Give your answer in standard form.

(3)

Worked solution

January 2021 | Question 6 | 3 marks

Family: Calculate and model in standard form • Variant: Add or subtract standard-form numbers

Method / working

1. Rewrite the terms with a common power of ten.
2. Add the coefficients to obtain $32 \times 10^{1^2}$ ■.
3. Normalize standard form: $3.2 \times 10^{1^2}$ ■.

Final answer: $3.2 \times 10^{1^2}$ ■

8 The table shows the populations of five countries.

Country	Population
China	1.4×10^9
Germany	8.2×10^7
Sweden	9.9×10^6
Fiji	9.1×10^5
Malta	4.3×10^5

- (a) Work out the difference between the population of China and the population of Germany.
Give your answer in standard form.

.....
(2)

Worked solution

June 2021 | Question 8 | 2 marks

Family: Calculate and model in standard form • Variant: Add or subtract standard-form numbers

Method / working

1. Subtract in matching powers: $1.4 \times 10^3 - 8.2 \times 10^2 = 1.318 \times 10^3$.
2. Give the result in standard form.

Final answer: 1.318×10^3

Given that

$$\text{population of Fiji} = \frac{1}{k} \times \text{population of Sweden}$$

(b) work out the value of k .

Give your answer correct to the nearest whole number.

$k = \dots\dots\dots$
(2)

Worked solution

June 2021 | Question 8 | 2 marks

Family: Calculate and model in standard form • Variant: Multi-step contextual standard-form problem

Method / working

1. Rearrange $k = (9.9 \times 10^4) / (9.1 \times 10^3)$.

2. Evaluate and round to the nearest whole number: $k \sim 11$.

Final answer: $k = 11$

- 9 A rainwater tank contains 2.4×10^7 raindrops.
The rainwater tank also contains 1.75×10^6 bacteria.

- (a) Work out the number of bacteria per raindrop in the tank.
Give your answer in standard form correct to 2 significant figures.

.....
(3)

A drop of rainwater contains 5.01×10^{21} atoms.

In a drop of rainwater the number of atoms is 3 times the number of molecules.

- (b) Work out the number of molecules in the rainwater tank.
Give your answer in standard form correct to one significant figure.

..... molecules
(2)

Worked solution

November 2021 | Question 9 | 3 marks

Family: Calculate and model in standard form | Variant: Multiply or divide standard-form numbers

METHOD / WORKING

1. For part (a), divide the number of bacteria by the number of raindrops:
2. $\frac{1.75 \times 10^6}{2.4 \times 10^7}$
3. $= \frac{1.75}{2.4} \times 10^{-1}$
4. $= 0.072916\dots$
5. Correct to 2 significant figures:
6. 7.3×10^{-2}
7. For part (b), one drop contains 5.01×10^{21} atoms.
8. The number of molecules in one drop is
9. $\frac{5.01 \times 10^{21}}{3} = 1.67 \times 10^{21}$
10. There are 2.4×10^7 drops, so the number of molecules in the tank is
11. $(1.67 \times 10^{21})(2.4 \times 10^7)$
12. $= 4.008 \times 10^{28}$
13. Correct to one significant figure:
14. 4×10^{28}
15. Answer: (a) 7.3×10^{-2} , (b) 4×10^{28} molecules

FINAL ANSWER

(a) 7.3×10^{-2}

- 9 A rainwater tank contains 2.4×10^7 raindrops.
The rainwater tank also contains 1.75×10^6 bacteria.

- (a) Work out the number of bacteria per raindrop in the tank.
Give your answer in standard form correct to 2 significant figures.

.....
(3)

A drop of rainwater contains 5.01×10^{21} atoms.

In a drop of rainwater the number of atoms is 3 times the number of molecules.

- (b) Work out the number of molecules in the rainwater tank.
Give your answer in standard form correct to one significant figure.

..... molecules
(2)

Worked solution

November 2021 | Question 9 | 2 marks

Family: Calculate and model in standard form | Variant: Multi-step contextual standard-form problem

METHOD / WORKING

1. For part (a), divide the number of bacteria by the number of raindrops:
2. $\frac{1.75 \times 10^6}{2.4 \times 10^7}$
3. $= \frac{1.75}{2.4} \times 10^{-1}$
4. $= 0.072916...$
5. Correct to 2 significant figures:
6. 7.3×10^{-2}
7. For part (b), one drop contains 5.01×10^{21} atoms.
8. The number of molecules in one drop is
9. $\frac{5.01 \times 10^{21}}{3} = 1.67 \times 10^{21}$
10. There are 2.4×10^7 drops, so the number of molecules in the tank is
11. $(1.67 \times 10^{21})(2.4 \times 10^7)$
12. $= 4.008 \times 10^{28}$
13. Correct to one significant figure:
14. 4×10^{28}
15. Answer: (a) 7.3×10^{-2} , (b) 4×10^{28} molecules

FINAL ANSWER

(b) 4×10^{28} molecules

5 (a) Write 5.7×10^{-3} as an ordinary number.

.....
(1)

(b) Write 800 000 in standard form.

.....
(1)

(c) Work out $\frac{3 \times 10^5 - 2.7 \times 10^4}{6 \times 10^{-2}}$

.....
(2)

.....
(Total for Question 5 is 4 marks)

Answer reference | May-Nov 2020 | Question 5

Family: Convert and compare standard form • Variant: Ordinary number to standard form

5	(a)		0.0057	1	B1
	(b)		8×10^5	1	B1
	(c)	$\frac{273000}{6 \times 10^{-2}}$		2	M1 for 273 000 or digits 455
			4 550 000		A1 for 4 550 000 or 4.55×10^6 oe
					Total 4 marks

Worked solution

May-Nov 2020 | Question 5 | 4 marks

Family: Convert and compare standard form | Variant: Ordinary number to standard form

METHOD / WORKING

Worked solution
May-Nov 2020 Question 5 4 marks
Family: Convert and compare standard form Variant: Ordinary number to standard form
Method
$5.7 \times 10^{-3} = 0.0057$
$800000 = 8 \times 10^5$
For part (c), first simplify the numerator:
$3 \times 10^5 - 2.7 \times 10^4 = 300000 - 27000 = 273000$
Then
$\frac{273000}{6 \times 10^{-2}} = (273000)/(0.06) = 4550000$
$4550000 = 4.55 \times 10^6$
Answer: (a) 0.0057, (b) 8×10^5 , (c) 4.55×10^6

FINAL ANSWER

- 9 (a) Write 6.25×10^{-4} as an ordinary number.

.....
(1)

- (b) Work out $(2.4 \times 10^{12}) \div (9.6 \times 10^4)$
Give your answer in standard form.

.....
(2)

(Total for Question 9 is 3 marks)

Answer reference | January 2023 | Question 9

Family: Convert and compare standard form • Variant: Ordinary number to standard form

9	(a)		0.000 625	1	B1
	(b)	25 000 000 or e.g. 25×10^6 or 0.25×10^8 or 2.5×10^n $n \neq 7$		2	M1
		<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	2.5×10^7		A1
					Total 3 marks

Worked solution

January 2023 | Question 9 | 3 marks

Family: Convert and compare standard form • Variant: Ordinary number to standard form

Official final mark-scheme pages are embedded immediately after this question.

8 (a) Write 5.76×10^4 as an ordinary number.

.....
(1)

(b) Work out $\frac{3 \times 10^5 + 8 \times 10^3}{4 \times 10^{-2}}$

Give your answer in standard form.

.....
(2)

(Total for Question 8 is 3 marks)

Worked solution

November 2025 | Question 8 | 3 marks

Family: Convert and compare standard form | Variant: Ordinary number to standard form

METHOD / WORKING

Worked solution
November 2025 Question 8 3 marks
Family: Convert and compare standard form Variant: Ordinary number to standard form
Method
$5.76 \times 10^4 = 57600$
For part (b), simplify the numerator:
$3 \times 10^5 + 8 \times 10^3 = 300000 + 8000 = 308000$
Then
$\frac{308000}{4 \times 10^{-2}} = (308000)/(0.04) = 7700000$
In standard form:
$7700000 = 7.7 \times 10^6$
Answer: (a) 57600, (b) 7.7×10^6
PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER